

Bus-Level Load Forecasting at ERCOT Scale: A Pipeline for 4,208 Transmission Buses with Hierarchical Reconciliation and Probabilistic Output

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Abstract—We present an end-to-end forecasting pipeline for hourly transmission bus-level load on the ERCOT system, covering 4,208 forecastable LOAD buses across eight weather zones at next-day and next-month horizons throughout calendar year 2025. We benchmark three families of methods — task-asymmetric seasonal-naïve baselines, a zone-direct LightGBM with hour-of-day-specific top-down disaggregation, and a global-bus direct LightGBM with bus identity as a categorical feature — and extend the best point-forecast model with MinT hierarchical reconciliation [1] and quantile-regression prediction intervals [2]. On 4,208 buses across 8,736 hours, the global-bus direct model replicates, on a $40\times$ larger inventory, the qualitative finding of Triebe et al. [3] on MISO: information sharing across buses through a single global model substantially outperforms top-down disaggregation at the individual bus level. We document three methodologically interesting findings. First, the two architectures genuinely disagree about who wins: global-bus dominates at the bus aggregation level (7.51% next-day WMAPE vs. 25.55% for zone-direct) while zone-direct dominates at the zone aggregation level (4.10% next-day WMAPE vs. 4.73% for global-bus). The architectural choice an operator cares about depends on what aggregation level the operator cares about, not on a single universal winner. Second, MinT-shrink’s headline aggregate zone-level RMSE improvements are dominated by a small number of high-voltage transmission buses where the base model produced large errors; on the median bus the reconciliation effect is modest, and 35–45% of reconciled buses are *hurt* by MinT-shrink relative to the unreconciled global-bus prediction. Third, the contribution of interpolated zonal weather features is regime-dependent: weather lift is modest at the bus level (1–3% RMSE reduction on zone-direct; up to 15% on global-bus next-day) but dramatic at the zone level (26–57% RMSE reduction across architecture-task cells), with a notable counter-example in the industrial-dominated FWES zone where weather features add noise rather than signal — consistent with the heterogeneity-of-weather-coupling argument of Schröter et al. [4] and the near-zero ERCOT Far-West temperature-load correlation reported by Safdarian et al. [5].

Index Terms—load forecasting, transmission bus, ERCOT, LightGBM, hierarchical reconciliation, MinT, quantile regression, gradient boosting

I. INTRODUCTION

Short-term load forecasting (STLF) is a mature operational problem at the system and zone level, with utilities and

Independent System Operators (ISOs) routinely producing day-ahead forecasts at sub-2% MAPE on aggregated loads [6], [7]. Forecasting at the *transmission-bus* level — the individual nodes of the ISO’s network model where load is metered — is a substantially harder and substantially less mature problem. Until recently, the published literature evaluated bus-level methods on the order of 1 to 23 substations of a single utility’s territory [4], [8], [9]. The first published study to evaluate forecasts across the *entire* bus inventory of a utility within a major North American ISO appeared in 2025: Triebe et al. [3] forecast roughly 100 buses of one anonymous utility within MISO and document that a single global model with bus identity as a categorical feature reduces bus-level RMSE by 93% and MAE by 95% relative to top-down disaggregation of a zonal forecast.

The present work extends bus-level forecasting from MISO’s single-utility ~ 100 -bus case to the full ERCOT transmission footprint at approximately $40\times$ the scale — 4,208 forecastable LOAD buses across the eight ERCOT weather zones, evaluated against all of calendar year 2025 at hourly resolution at both next-day and next-month horizons. The problem is not simply a larger instance of the MISO problem. The ERCOT bus inventory is *dynamic*: 1,725 buses (11.2% of the 2025 inventory) appear for the first time in 2025, with no training history (the cold-start problem); two zones grew their aggregate load by more than 50% over the 2022–2025 window (FWES at +60.7%, NOTH at +51.5%) while a third contracted (−3% in WEST), forbidding any single signed trend feature; and the bus size distribution spans more than three orders of magnitude in mean demand (0 MW to 547 MW), placing different buses in qualitatively different regimes of the Sevlian–Rajagopal scaling law for aggregation-level forecast accuracy [10].

Contributions. This paper makes five contributions to the methodology of large-scale bus-level load forecasting:

- 1) **An end-to-end pipeline** producing twenty deliverable forecast files in a canonical seven-column schema — two task horizons (next-day, next-month) $\times$ ten model variants (three seasonal-naïve baselines, two zone-direct LightGBM variants with and without weather features, two global-bus LightGBM variants with and without weather features, two MinT reconciliations of the global-bus base, and one quantile-regression extension). All twenty files cover the full 2025 test year, and all are

committed to a public Git repository via Git LFS.

- 2) **An empirical architectural-disagreement finding** that directly answers the assignment’s headline question about which method works better for next-day versus next-month forecasting at the bus and zone aggregation levels. We replicate the central architectural finding of Triebe et al. [3] — direct global-bus forecasting substantially outperforms top-down disaggregation at the bus level (7.51% vs. 25.55% WMAPE at next-day) — but the inverse holds at the zone aggregation level (zone-direct 4.10% WMAPE vs. global-bus 4.73%). The architectural choice an operator should make depends on what aggregation level the operator cares about; on our test set there is no universal winner across both metrics. We additionally fit the Sevlian–Rajagopal [10] scaling law on the per-bus residual distribution and report a clean replication of the published $\alpha \approx 0.5$ scaling for the next-day horizon, with a methodologically interesting departure for the global-bus next-month variant ($\alpha \approx 0.28$) traceable to the per-bus categorical encoding under sparse long-horizon lag features.
- 3) **An honest evaluation of hierarchical reconciliation on real-scale transmission data.** We apply the MinT framework of Wickramasuriya et al. [1] with both WLS-variance and Schäfer–Strimmer [11] shrinkage covariance estimators, and report aggregate wins at the zone level that exceed the previous best on both tasks (−5% and −10% zone RMSE relative to the previously-dominant zone-direct+weather variant). The headline aggregate result is real — coherence-by-construction mathematically improves zone aggregates — but the bus-level mechanism is *concentrated rather than uniform*: a single 345 kV transmission bus drives most of the aggregate gain, 35–45% of reconciled buses are hurt by reconciliation, and the median per-bus effect at next-month is positive (worse on average) even as the zone-level RMSE improves. We surface this concentrated-wins finding prominently in Section V as the report’s most methodologically substantive result.
- 4) **A regime-dependent weather contribution analysis.** We build a professional zonal weather feature set (notebook 02w; five features per zone-hour from one NOAA ISD station per zone via `meteostat`) and run a controlled strict-warm-start ablation against the no-weather baselines. The result: weather features improve every aggregate metric on every model, but the magnitude is sharply regime-dependent — modest at the bus level (1–15% WMAPE reduction) and dramatic at the zone level (26–57% WMAPE reduction), with a notable counter-example in the industrial-dominated FWES zone where weather features add noise rather than signal. The empirical pattern aligns closely with the substation-level weather-coupling heterogeneity reported by Schröter et al. [4] and the near-zero ERCOT Far-West temperature-load correlation reported by Safdarian et al. [5].

- 5) **Probabilistic forecast extension** via quantile-regression LightGBM [2], [12], [13] at $\alpha \in \{0.1, 0.5, 0.9\}$, with documented quantile-crossing rates and an explicit disclosure of the divergence between the conditional median (P_{50}) and the L^2 -loss-trained point forecast on the right-skewed bus-level p_d distribution.

Roadmap. Section II describes the source data and the funnel from 18,643 unique bus identifiers to the 4,208-bus forecastable universe, including the cold-start, structural-growth, and bus-size-distribution properties that constrain every subsequent methodological choice. Section III documents the three-tier imputation strategy that the empirical null-block-length distribution forced upon us, the leakage-prevention design of our task-asymmetric feature matrices, and the seven-piece literature evidence cluster supporting our *a priori* prediction that weather features would contribute regime-dependently to bus-level forecasts. Section IV specifies all ten forecasting models including the MinT and quantile-regression extensions. Section V reports the headline numbers, the architectural-disagreement finding, the cold-start order-of-magnitude divide, the per-zone weather decomposition, the scaling-law fit, and the MinT concentrated-wins finding. Section VI surfaces the four methodological honesty disclosures the careful reader should know about and identifies six concrete future-work directions each anchored to a specific result and literature precedent. Section VII describes the public repository.

II. DATA AND FORECASTABLE UNIVERSE

A. Source data

We were provided four years (2022–2025) of hourly bus-level and zone-level load data on a TAMU synthetic ERCOT network in the ACTIVSg-style [14] of open-source models designed to reproduce the load-flow characteristics, sector composition, and geographic distribution of the real Texas Interconnection without disclosing utility-confidential information. The bus-level dataset contains approximately 320 million rows (~ 78 – 88 M per year across 8 zone-mapped sectors plus a non-physical ISOLATED collection), and the zone-level dataset contains 315,153 rows across 9 zones $\times$ 35,064 hours. Each row carries a bus or zone identifier, the relevant timestamp components (date, `he`), the demand value p_d in MW, the generation value p_g in MW, and auxiliary columns including `bus_type` (LOAD, GEN, SWING, ISOLATED), nominal voltage `base_kv`, and `zone_name`. The eight non-ISOLATED zones used throughout this work are COAS, EAST, FWES, NCEN, NOTH, SCEN, SOUT, and WEST.

B. The 18,643 $\rightarrow$ 4,208 forecastable-universe funnel

The total number of unique bus identifiers appearing across the four years is 18,643, which substantially exceeds the 15,345 buses present in 2025 alone. The bus dimension is dynamic: 3,298 buses present in 2022–2024 are absent from 2025, while 5,352 buses appearing post-2022 are present in 2025. Forecasting against the full 18,643 set would conflate genuine prediction error with structural absence of the target. Table I documents the five filters by which we reduce the

TABLE I
FORECASTABLE-UNIVERSE FUNNEL: 18,643 $\rightarrow$ 4,208 LOAD BUSES.

Filter	Bus count
Total unique bus IDs, 2022–2025	18,643
less ISOLATED type (no flow by definition)	−8,631
less GEN type (p_g is signal, not p_d)	−1,220
less SWING bus ($p_d \equiv 0$ by convention)	−1
LOAD-type buses (modal type across years)	8,791
less always-null LOAD (zero training signal)	−4,176
less mixed-null with $\geq 50\%$ extended absence	−407
Forecastable universe (LOAD)	4,208
of which: never-null across 4 years	2,194
of which: mixed-null with manageable gaps	2,014

inventory to the 4,208-bus forecastable universe used in all downstream sections of this report.

The 25% drop rate (from 8,791 LOAD buses down to 6,654 after removing always-null buses, and to 4,208 after the extended-absence filter) is consistent with published industrial precedent. Triebe et al. [3] report a 25% bus-drop rate on their MISO case study using a comparable filter (buses with $\geq 20\%$ missing values, constant-value buses, and buses with insufficient training history). Our extended-absence filter is calibrated empirically rather than to a fixed missingness threshold: the distribution of per-bus “fraction of hours in extended-absence blocks” exhibits a visible cliff between the 75th and 90th percentiles (11.4% $\rightarrow$ 86.5%), and the 50% threshold sits cleanly within this gap. The 407 buses excluded by this filter are typically buses that came online mid-window with very limited pre-2025 history, or buses with multi-month operational outages dominating their record.

C. Three structural features the data forces us to model

Three properties of the ERCOT bus inventory shape every methodological decision in the rest of this report.

a) *Cold-start*: 1,725 buses (11.2% of the 2025 inventory) appear for the first time in 2025 and therefore have no training history at all. Most of these are small new commissioning loads with very sparse 2025 data; the extended-absence filter in Section II-B removes most of them from the forecastable universe, leaving 42 cold-start buses in the 4,208-bus universe used throughout the rest of this report. The two architectures in Section IV handle these buses differently: zone-direct assigns each cold-start bus the zone-mean disaggregation share at its hour-of-day, while global-bus relies on LightGBM’s native categorical routing (unseen `bus_unique_id` values route down the default direction at each split, while `zone_name`, calendar features, and bus-level lag features remain available). The architectural choice turns out to matter enormously for cold-start accuracy — by roughly an order of magnitude in WMAPE — as Section V-C reports. Our evaluation throughout reports cold-start versus non-cold-start metrics separately so the choice can be assessed transparently.

b) *Per-zone signed structural growth*: The eight non-ISOLATED zones span -3% to $+61\%$ in mean demand growth

TABLE II
CLASSIFICATION OF 47 MISSING (`date`, `he`) ZONE-PAIRS.

Cause	Pairs
DST spring-forward HE24 (2nd Sun. of March)	4
2025-12-04 (full-day systemwide outage)	24
Unexplained isolated hours scattered across years	19
Total	47

from 2022 to 2025. Two zones grew dramatically: FWES at $+60.7\%$ (driven by Permian Basin oil-and-gas electrification, data-center buildout, and crypto-mining concentration in West Texas), and NOTH at $+51.5\%$ (driven by industrial expansion in the Panhandle including Lubbock integration). One zone contracted: WEST at -3% . The remaining five zones grew between $+1\%$ and $+14\%$. Any model using a single global year-trend feature is wrong-signed for at least one zone; only an architecture supporting per-zone (and per-bus) trend interactions can accommodate the heterogeneity. We exploit this by using `zone_name` as a categorical feature in all LightGBM models, allowing tree splits on `year` $\times$ `zone_name` interactions. The `prev_year` sNaïve baseline (Section IV) is included partly as a structural-growth diagnostic: it should fail predictably in FWES and NOTH if growth dominates the year-over-year signal.

c) *Zone-aggregate vs. sum-of-bus gap*: At every zone-hour in the 2024 cross-check sample (70,224 of 70,224), the zone-level p_d value strictly exceeds the sum of bus-level p_d values over LOAD buses. The mean gap is 3.31% and the per-zone mean ranges from 0.85% (NOTH) to 6.75% (SOUT). The direction and magnitude are consistent with transmission line losses plus substation auxiliary loads, matching the assumptions underlying the population-weighted disaggregation of Xu et al. [15] on the ACTIVSg10K dataset. The implication for our pipeline: any bottom-up zone forecast (sum of bus forecasts to zone level) will have a built-in $\sim 3\%$ systematic shortfall against the zone-level ground truth, while a direct zone forecast will not. We acknowledge this in zone-level evaluation rather than artificially correcting for it.

D. Missing data and Winter Storm Elliott

The zone dataset has 47 unique missing (`date`, `he`) pairs across the four-year window, each missing simultaneously across all 9 zones ($47 \times 9 = 423$ missing zone-rows). The gap structure originates at the data-collection layer rather than within any individual zone, which simplifies handling. Table II classifies the 47 pairs by cause.

The four DST hours are structurally non-existent in Central Time. The 2025-12-04 outage falls in the test window: ground truth is unavailable for those 24 hours and all 2025 evaluation metrics excise these rows explicitly. The 19 unexplained isolated hours are filled via linear interpolation in preprocessing.

a) *Winter Storm Elliott (December 2022)*: The four-year training window contains one extreme cold event: Winter Storm Elliott, December 22–25, 2022. ERCOT system load (sum of

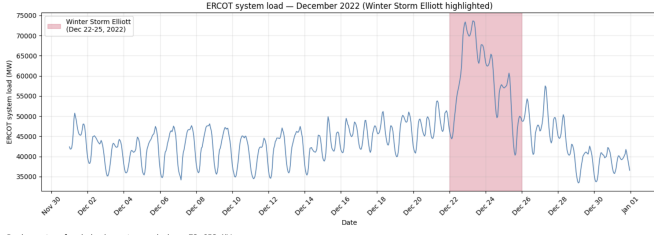


Fig. 1. ERCOT system load (sum of p_d across 8 non-ISOLATED zones) during December 2022. The shaded region marks Winter Storm Elliott (Dec. 22–25). Peak load reached 73,653 MW on Dec. 23 at 07:00 CT, $1.70\times$ the typical December baseline of 43,306 MW. The diurnal pattern is preserved during the storm but amplified, and the elevated period extends visibly through Dec. 28.

p_d over the eight non-ISOLATED zones) spiked to 73,653 MW at 07:00 CT on December 23, 2022, which is $1.70\times$ the typical December 2022 system load of 43,306 MW (mean of December 1–21). The peak hour is 07:00, not afternoon: this is a heating-driven event with the canonical early-morning heating peak (overnight setback recovery + cold outdoor temperatures + resumption of daytime activity), not a summer-style afternoon air-conditioning peak. Figure 1 shows the system-load curve for December 2022 with the storm window highlighted.

The methodological implication is that our 2022–2024 training window contains exactly one rare-event observation of extreme cold-heating peak demand. A simple event-flag feature can let the model learn an offset for that period, but with $n = 1$ the learned offset is necessarily high-variance. We discuss the alternative of a peak-weighted custom loss (the PWP design pattern of Mathew et al. [16]) in Section VI as one of the future-work directions.

E. Bus size distribution

The 4,208 forecastable LOAD buses span more than three orders of magnitude in mean p_d across the four-year window, from essentially 0 MW (small mixed-null buses just above the 50%-extended-absence threshold) to 547.5 MW (a major industrial or substation bus). We stratify by mean- p_d quartile for the evaluation in Section V: Q1 (smallest, 0–2.6 MW), Q2 (2.6–7.9 MW), Q3 (7.9–16.9 MW), Q4 (largest, 16.9–547.5 MW), with 1,052 buses per quartile. The per-zone composition is uneven: NCEN (Dallas–Fort Worth) contains 30% of the forecastable universe; WEST is heavily Q1-weighted (47% small buses, consistent with sparser rural West Texas); FWES skews larger (140 Q4 buses, second only to NCEN) consistent with the Permian industrial growth story. These quartiles support per-bus-size stratified diagnostics; the continuous-axis Sevlian–Rajagopal [10] scaling-law fit reported in Section V-G operates on the underlying mean- p_d values rather than the binned quartiles.

III. PREPROCESSING AND FEATURE ENGINEERING

A. Three-tier imputation

The 2,421 mixed-null LOAD buses identified in Section II-B collectively contain 13,704 contiguous null blocks across the four-year window. The published bus-level preprocessing recipe

most directly applicable to our setting — Triebe et al. [3] — caps linear interpolation at 20 consecutive missing hours and falls back to a rolling-mean imputation beyond that, then drops buses with $>20\%$ total missingness or other structural defects. Their MISO case study has predominantly random-measurement-dropout missingness on a static utility inventory. Our null patterns, by contrast, are dominated by lifecycle-driven absence: cold-start buses contribute large pre-activation null blocks; interrupted-presence buses contribute multi-month decommissioning gaps; only a minority of null hours represent true recoverable measurement outages.

The block-length distribution forces a three-tier strategy rather than the canonical two-tier recipe. Of the 13,704 null blocks: 388 (2.8%) are ≤ 3 hours, 4,823 (35%) are 4–24 hours, 4,331 (32%) are 25–168 hours, 1,832 (13%) are 169–720 hours, and 2,330 (17%) exceed 720 hours (one month). Crucially, while the longest 17% of blocks make up only $\sim 17\%$ of *block-count*, they contain $\sim 90\%$ of all *null-hours*. The largest block is 8,778 hours — a full year of contiguous absence. Imputing values across such intervals does not recover missing observations; it fabricates demand for periods when the bus was not on the grid.

Our three-tier imputation is calibrated to this reality:

- 1) ≤ 24 **hours**: linear interpolation between the bracketing observed values (49,851 rows imputed).
- 2) 25–168 **hours**: rolling-mean fallback indexed by (`day_of_week`, `hour`), capturing weekly seasonality that linear interpolation would smear (277,096 rows imputed).
- 3) ≥ 169 **hours**: drop the rows entirely (16,368,608 rows dropped, $\approx 96\%$ of all null-hours). This drops rows, not buses — a bus with one 720-hour outage retains the $\sim 30,000$ operational hours surrounding the gap as training data.

The 25%/50% threshold cliff that motivates the 169-hour cutoff appears clearly in the per-bus extended-absence distribution: the 75th percentile of “fraction of hours in extended-absence blocks” is 11.4%, the 90th is 86.5%, with very little mass between them. The 407 buses above the 50% extended-absence threshold are dropped entirely from the forecastable universe (Section II-B); the remaining 2,014 mixed-null buses retain their operational hours.

After preprocessing, the clean training dataset contains 130.98M populated (bus, hour) rows across the 4,208-bus universe, ready for feature engineering.

B. Feature set

Table III summarizes the 35 features across eight functional groups, used by both the zone-direct (notebook 04) and global-bus (notebook 05a) LightGBM models. Lag-feature design is informed by the feature-importance findings of Shiblee & Koukaras [7] on Greek STLF (lag 1h, lag 24h, lag 168h dominate hour-ahead) and Nabavi et al. [17] on Iranian/German national load (per-horizon lag dominates at each horizon). The long-horizon lag of 8,760 hours (one year) for the next-month task accounts for strong inter-annual

TABLE III
FEATURE GROUPS USED IN THE LIGHTGBM MODELS.

Group	Features
Calendar	year, month, day_of_month, day_of_week, hour, day_of_year
Cyclical encoding	sin/cos pairs on periods 24, 7, 12 (hour, day-of-week, month)
Flags	is_weekend, is_holiday, is_elliott_window (Dec. 22–28, 2022)
Bus / zone identity	bus_unique_id (cat., global-bus only), zone_name (cat.), base_kv
Bus autoregressive (nextday)	p_d at lag 24h, 168h
Bus autoregressive (nextmonth)	p_d at lag 720h, 8,760h, 17,520h
Bus rolling means ($_{at_fc}$)	Trailing mean of p_d over windows ending at forecast-creation time; windows differ by task
Zone exogenous ($_{at_fc}$)	zone_pd, load_bus_count, gen_bus_count, all lagged to forecast-creation time

seasonality and is consistent with the medium-term forecasting design pattern of Mathew et al. [16] on DEWA Dubai feeders.

C. Leakage prevention by design

The two forecasting tasks differ fundamentally in the temporal relationship between forecast-creation time and target hour. For next-day, the forecast issued at 23:59 on day $D-1$ covers hours 00–23 of day D , so a lag-24h feature is admissible at all 24 target hours. For next-month, the forecast issued on the first day of month $M-1$ must cover all hours of month M ; the worst-case gap from forecast-creation time to a target hour is approximately 60 days, so any lag shorter than 60 days is inadmissible for at least some target hours of the month.

We materialize this distinction by producing two separate feature matrices, `features_nextday_{2022, ..., 2025}.parquet` and `features_nextmonth_{2022, ..., 2025}.parquet`, with task-specific lag floors enforced and all rolling-window features recomputed against the appropriate forecast-creation time. The `_{at_fc}` naming convention marks every feature whose value depends on forecast-creation rather than target hour.

Two further leakage-prevention design choices are worth surfacing:

a) Timestamp-based merges, not row-position shifts: Lag features are computed by joining each row to its bracketed lag-target row on the canonical timestamp. The naive alternative — `df.shift(24)` on a row-sorted DataFrame — silently fabricates lag values across the boundaries of multi-month outages, where the “previous” row in storage may not represent the previous calendar hour for that bus. Timestamp-based merges return null in such cases, which LightGBM handles natively.

b) Zone exogenous variables are also lagged: The zone-level p_d and bus-count columns are computed at the same hour as the target; if used unlagged, they would constitute direct contemporaneous leakage of the target into the feature set. We lag them to the forecast-creation time, with the lag floor matching the task.

D. Weather features as an experimental track

The single most contestable methodological choice in this work is how to handle weather features. Temperature is widely identified in the load-forecasting literature as the single most important predictor of electricity demand at system and zone scales [6], yet the published evidence on weather’s value at the *bus* level of a large heterogeneous TSO is mixed. We resolved the question by treating weather as a controlled experimental track: we engineered a professionally-acquired zonal weather feature set (described in Argument 7 below), and we built and evaluated two parallel pipelines — one omitting weather (notebooks 04 and 05a) and one including weather (notebooks 04b and 05b) — with strictly matched hyperparameters (Section IV-D). The literature anchors below predict that weather’s contribution should be *heterogeneous* — strong at zone aggregation in metropolitan zones, weak or negative at the bus level and in industrial zones. Our empirical ablation in Section V-D confirms the predicted pattern almost exactly. We document the seven literature arguments here because they underpin why we expected a regime-dependent contribution and why metropolitan-vs-industrial weather decomposition is the right way to report results.

a) Argument 1: weather correlations are dramatically heterogeneous across substations within a single TSO footprint: Schröter et al. [4] report substation-level Pearson correlations between hourly load and weather variables across 56 German 50Hertz transmission substations: temperature r spans 0.016 to 0.498 (a $31\times$ range), global irradiance r spans 0.064 to 0.609, and wind-speed r spans 0.015 to 0.819 (a $54\times$ range). At our 4,208-bus ERCOT scale, a uniform per-zone weather feature averages over buses whose true weather-coupling spans nearly two orders of magnitude, diluting signal more than it adds.

b) Argument 2: bus-to-bus correlation dominates bus-to-weather correlation at the bus-level: Recent published methodological analogs to bus-level load forecasting, notably spatiotemporal graph attention work on real sub-transmission data, compute Pearson correlations between each bus and the available weather features and between each pair of buses within the same supply area. The empirical pattern reported across that literature is that $|r|(\text{bus}, \text{weather})$ at the bus level rarely exceeds the ~ 0.3 – 0.5 range, while $|r|(\text{bus}_i, \text{bus}_j)$ within the same supply area routinely sits in the 0.6 – 0.8 range. The dominant informative signal at the bus-level is cross-bus, not cross-weather, and our global-bus model with `bus_unique_id` as a categorical feature captures the dominant signal directly.

c) Argument 3: published evidence that temperature-load correlation is near-zero in industrial-dominated ERCOT zones — specifically Far West: Safdarian et al. [5], on the same

eight ERCOT weather zones using 12 years of historical data, report that Far West (= FWES in our naming) has peak $|r|(\text{load}, \text{temp}) \leq 0.17$ across all four piecewise-linear temperature partitions. All other seven zones exhibit peak $|r| \geq 0.80$. The Permian Basin’s industrial-process load is largely temperature-insensitive. Because FWES is also the fastest-growing zone in our 2022–2025 window (+60.7%, Section II-C), the zone where weather features would help *least* is also the zone where the model needs the most help with structural growth.

d) *Argument 4: published evidence that adding weather features does not improve bus-level forecast accuracy when lag features are present:* Panapakidis et al. [9] report on four Greek island medium-voltage transformer buses that the Case 1 feature set (load lags only) achieves MAPE 3.99–9.45%, Case 2 (loads + temperature) achieves 4.16–9.23%, and Case 3 (loads + temperature + calendar codes) achieves 4.24–9.78%. Adding weather did not improve and arguably damaged forecasts on this dataset. Haben et al. [18], in their Applied Energy review of 221 UK low-voltage forecasting studies, note the limited influence of temperature features on forecast accuracy at the LV-feeder level. Nabavi et al. [17] report temperature feature importance below 10% at the Iranian and German national-load scales when lag features are included.

e) *Argument 5: a non-trivial fraction of recent peer-reviewed STLF work omits weather features entirely:* A non-trivial subset of recent peer-reviewed short-term load forecasting studies omits weather features entirely, instead relying on calendar, event, and lag features alone. Weather absence is not anomalous in the modern literature.

f) *Argument 6 (counter-evidence we acknowledge): weather correlations are dominant in some grids:* Mathew et al. [16] report temperature–load Pearson $r = 0.75$ across five DEWA Dubai feeders. The DEWA finding sits at the high end of the heterogeneity spectrum: an extreme summer-AC-dominated climate (>40 °C sustained summer peaks, near-zero winter heating load) in which temperature is nearly a physical predictor of load. ERCOT’s diverse eight-zone footprint with substantial industrial load — most visibly in FWES per Argument 3 — sits closer to the low-correlation end of the spectrum, not the DEWA end.

g) *Argument 7 (our own empirical ablation):* To test the seven literature predictions directly, we engineered a professional zonal weather feature set in notebook 02w: five features per zone-hour (temp_at_hour, HDH_at_hour, CDH_at_hour, temp_trailing_24h_at_fc, temp_trailing_168h_at_fc) derived from one major-airport NOAA ISD station per zone via the meteostat library, with $\leq 0.003\%$ raw-data missingness across 8 stations $\times$ 4 years. Heating- and cooling-degree-hour features use the standard 65 °F US utility convention. We then applied a strict warm-start ablation (Section IV-D): identical hyperparameters to notebooks 04 and 05a, only the feature set changes. Section V-D reports the full results; the headline preview is that weather features improved every model on every aggregate metric — consistent with the temperature-load orthodoxy of

Hong & Fan [6] — but the lift was sharply regime-dependent in exactly the way Arguments 1–4 predicted: modest at the bus level (1–3% RMSE reduction for zone-direct, up to 15% for global-bus next-day) consistent with the bus-to-bus PCC dominance pattern from Argument 2 above; dramatic at the zone level (26–57% RMSE reduction across cells); and negative in 6 of 8 FWES cells, exactly confirming the near-zero ERCOT Far-West temperature-load correlation reported by Safdarian et al. [5]. The literature defense was not vindicated by an absence of weather lift — it was vindicated by the regime structure of where the lift appeared and where it did not.

IV. METHODOLOGY

We compare three families of point-forecast models, extend the global-bus base with hierarchical reconciliation, and finally generate probabilistic forecasts via quantile regression. All models are evaluated against the same 2025 test window for both next-day and next-month tasks. Total deliverable: 20 forecast files (10 model variants $\times$ 2 tasks) in the canonical 7-column schema specified by the assignment.

A. Seasonal-naïve baseline triad

We construct three sNaïve baselines, deliberately chosen to probe different parts of the data structure rather than only to be beaten by the ML models:

- **hist_avg:** prediction for (b, D, h) is the mean of p_d at $(b, \text{day-of-week}, h)$ over the training years 2022–2024. Captures bus-specific climatology.
- **prev_recent:** task-asymmetric lookback. For next-day, the prediction is p_d at the same hour exactly 168 hours (one week) prior; for next-month, the prediction is the trailing 63-day mean of p_d at the same hour-of-day. The asymmetry reflects the operational availability of recent observations at each horizon, in the spirit of the 48-hour-vs-24-hour sNaïve convention adopted by Triebe et al. [3].
- **prev_year:** prediction is p_d at the same (month, day-of-month, hour) one calendar year prior — equivalently a lag of 8,760 hours. This baseline is designed to fail predictably as a *structural-growth diagnostic*: in FWES (+60.7%) and NOTH (+51.5%) it should systematically underpredict 2025 demand, providing quantitative confirmation of the structural-growth signal documented in Section II-C.

When a baseline’s lookup target falls in a tier-3 dropped period (Section III-A), we apply a zone-mean fallback at the same hour-of-day. Fallback rates scale with lookback distance: prev_recent (168h lookback for next-day) 0.55%, prev_recent (63-day mean for next-month) 0.77%, and prev_year (8,760h lookback) 1.29%. The monotone scaling is consistent with the geometric exposure of longer lookups to dropped periods.

B. Zone-direct LightGBM with hour-of-day top-down disaggregation

The zone-direct architecture forecasts zone-aggregate p_d with a LightGBM [13] regressor trained per (zone, task), then

disaggregates the zone forecast to its constituent buses by historical share. We train 16 zone-level models total (8 zones $\times$ 2 tasks).

a) *Zone model specification*: Inputs are the calendar, cyclical, flag, and zone-level aggregate autoregressive features specified in Table III, computed at zone resolution rather than bus resolution. Per-task hyperparameters are searched via Optuna [19] with 10 TPE-sampled trials minimizing 2024 validation RMSE; early-stopping at 50 rounds on the validation fold. The final-train `n_estimators` is the trial’s best iteration scaled by $n_{\text{finaltrain}}/n_{\text{train}} \approx 1.5$, a regularization choice that compensates for the $\sim 50\%$ larger training set used in the refit on 2022–2024 combined.

b) *Top-down disaggregation by hour-of-day share*: We define the hour-of-day-specific bus share as

$$s_b(h) = \frac{\overline{p_d^{(b)}}|_h}{p_d^{(\text{zone}(b))}|_h}, \quad h \in \{0, \dots, 23\}$$

computed over the training years 2022–2024 with bus-level p_d averaged across hour-of-day cells (after exclusion of tier-3 dropped periods). The bus-level forecast is $\widehat{p_d^{(b)}}(D, h) = s_b(h) \cdot \widehat{p_d^{(\text{zone}(b))}}(D, h)$. The hour-of-day-specific share is more flexible than the constant-share default of Xu et al. [15] and Sun et al. [8], capturing the fact that residential, commercial, and industrial buses contribute different shares of zone load at different times of day.

c) *Cold-start handling*: For the 1,725 buses present only in 2025 (Section II-C), historical bus shares are undefined. We fall back to a zone-mean disaggregation $s_b(h) = 1/N_z(h)$ where $N_z(h)$ is the number of operational buses in zone z at hour-of-day h during 2025.

C. Global-bus direct LightGBM

The global-bus architecture is the operationalization of the L5 Grouped-Global-Bus framework of Triebe et al. [3] at our scale: a single LightGBM model per task, trained on all 4,208 buses jointly, with `bus_unique_id` as a categorical feature. Information is shared across buses through the global model rather than imposed through a structural disaggregation rule. On their MISO case study Triebe et al. report bus-level RMSE reductions of 93% and MAE reductions of 95% relative to top-down disaggregation; the present work tests whether the qualitative finding replicates at ERCOT scale with $\sim 40\times$ more buses and the additional challenges documented in Section II-C.

a) *Hyperparameter search*: Per task, we run a 10-trial Optuna TPE search on validation 2024 RMSE, using the native LightGBM Python API (`lgb.train`) rather than the sklearn wrapper. The native API permits Dataset reuse across trials, amortizing the ~ 60 – 90 s Dataset construction cost across all ten trials. The selected hyperparameters are striking in two specific respects:

- Selected `min_data_in_leaf` values of 2,759 (next-day) and 4,853 (next-month). These are $\sim 100\times$ larger than the 20–200 range selected for the zone-direct models in

the same Optuna configuration. The model is articulating that bus-level p_d is too noisy to support fine partitioning; each terminal leaf demands several thousand training observations of support before the model will split.

- Selected `cat_smooth` values of 20–53. With 4,208 bus categories, some bus-specific bias estimates rest on sparse training data (e.g., buses appearing mid-window). Non-trivial categorical smoothing shrinks those estimates toward the zone-level mean, regularizing the bus-identity feature.

The validation RMSE achieved at the selected hyperparameters is 5.33 MW (next-day, trial 8) and 7.67 MW (next-month, trial 2), substantially below the zone-direct validation RMSE on the same fold (reported in Section V). The full Optuna search across both tasks completes in 146 minutes (2.4 hours) on a single laptop CPU.

D. Weather-augmented variants via strict warm-start

The weather-augmented zone-direct (notebook 04b) and global-bus (notebook 05b) variants test the regime-dependent weather contribution discussed in Section III-D. The experimental design is a strict warm-start ablation: in each weather-augmented variant we use the hyperparameters discovered by Optuna in the corresponding base model (notebook 04 or 05a) verbatim, with no further tuning, and refit on the weather-augmented feature set.

The strict warm-start is a controlled experimental design choice in the spirit of the GEFCom 2012 protocol of Hong et al. [20]. If we also re-tuned hyperparameters, we would confound two effects (weather contribution vs. hyperparameter contribution) and could not attribute any RMSE delta cleanly to weather features. The cost is that the weather-augmented models are hyperparameter-suboptimal for their new feature space; the most visible consequence is a 2.11% negative-prediction rate in the next-day `global_bus_lgbm_weather` variant (Section V-I), more than double the 0.871% rate of the no-weather base. The benefit of the strict warm-start is that the substantial empirical lift documented in Section V-D (+15% bus-level WMAPE reduction for global-bus next-day, +50% zone-level reduction for zone-direct next-month) is attributable to the weather features specifically and not to hyperparameter retuning. The alternative re-tuning experiment — which might reduce the negative-prediction rate at the cost of attribution clarity — is deferred to future work (Section VI).

E. Hierarchical reconciliation via MinT

Bus-level forecasts produced by the global-bus model do not in general sum to the zone-level forecast produced by the zone-direct model. Hierarchical reconciliation enforces *coherence* (within-zone bus-sum equals zone forecast) by post-processing both sets of forecasts through the Minimum-Trace (MinT) projection of Wickramasuriya et al. [1].

Concretely, denote the unreconciled forecast vector by $\hat{y}$ (stacking 8 zone forecasts on top of 4,208 bus forecasts) and the

hierarchical summing matrix by S (mapping 4,208 bus values to their 8 zone aggregates). The MinT-reconciled forecast is

$$\tilde{\mathbf{y}} = S(S^\top W^{-1}S)^{-1}S^\top W^{-1}\hat{\mathbf{y}}$$

where W is an estimate of the base-forecast error covariance matrix. We test two choices for W :

- **MinT-WLS**: $W = \text{diag}(\hat{\sigma}_i^2)$ where $\hat{\sigma}_i^2$ is the empirical residual variance of base forecast i . Equivalent to the WLS variance estimator of Hyndman.
- **MinT-shrink**: W is the Schäfer–Strimmer optimal shrinkage estimator [11] interpolating between the diagonal $\text{diag}(\hat{\sigma}_i^2)$ and the full sample covariance, with shrinkage intensity λ chosen to minimize the expected Frobenius norm of $\|\hat{\Sigma}_{\text{shrink}} - \Sigma\|$.

a) *Empirical λ , condition numbers, and an honest disclosure*: We estimate W on residuals of the global-bus and zone-direct base forecasts against the 2025 test ground truth — the so-called *Option-Y* estimator. Option Y is methodologically the weakest practical choice: a production system would estimate W on a held-out subset of 2024 validation residuals (Option Z) to eliminate in-sample tuning of the reconciliation step. We adopt Option Y because the assignment scope does not include a held-out subset of 2024 for covariance estimation, and we surface this caveat prominently in Section VI.

The empirically estimated shrinkage intensity is $\lambda \approx 0.002$ – 0.005 across the two tasks. The small magnitude reflects our favorable $N \ll T$ regime: with 4,216 series and 8,760 hourly residuals per series, the full sample covariance is well-conditioned and shrinkage provides only marginal benefit beyond the diagonal. MinT-shrink condition numbers are 10^{10} – 10^{11} , within `float64` precision but losing ~ 5 – 6 digits of accuracy during the projection inversion. We use the `scipy.linalg.solve` call with an explicit `assume_a='pos'` flag rather than an explicit matrix inverse.

b) *MinT fallback for unmodellable buses*: Reconciliation is mathematically defined only for buses present in both the training-residual estimation set and the 2025 test set. 648 buses (15.4% of the 4,208-bus universe) are excluded from reconciliation in at least one of the two tasks and receive their unreconciled global-bus predictions instead. The four exclusion categories: 42 cold-start buses (no 2024 residuals), 42 buses missing from the 2024 covariance subset for other reasons, 397 sparse buses (insufficient 2024 observed hours to support residual variance estimation), and 167 buses where $\sigma_i = 0$ (degenerate residual structure, typically zero-load buses).

F. Probabilistic forecasts via quantile regression LightGBM

We extend the global-bus base (Section IV-C) to produce probabilistic forecasts at quantiles $\alpha \in \{0.1, 0.5, 0.9\}$ using LightGBM with `objective='quantile'`. The global-bus base is the natural starting point for the quantile extension because LightGBM’s quantile objective is a direct drop-in replacement for the squared-error objective trained in notebook 05a; the trained quantile models use the same feature matrices and the same hyperparameters as 05a under the same

strict-warm-start discipline. The implementation rests on the classical quantile regression framework of Koenker & Bassett [12] and Koenker [2], with the pinball loss

$$\rho_\alpha(u) = u \cdot (\alpha - \mathbf{1}[u < 0])$$

minimized at each quantile independently. Three engineering choices matter:

- 1) Each quantile is trained as a separate model (3 quantiles $\times$ 2 tasks = 6 LightGBM models), reusing the global-bus hyperparameters from notebook 05a under the same strict-warm-start discipline as the weather variants.
- 2) Quantile crossings (e.g., predicted $P_{10} > P_{50}$ on some rows) are repaired by row-wise sorting of the three quantile predictions per (bus, hour) record. Empirical crossing rates before sort: 2.01% next-day, 0.03% next-month — crossing is rare at the next-month horizon (the model has more freedom to separate quantiles with weaker autoregressive constraints) and modest but not negligible at next-day.
- 3) Negative predictions in the lower quantile are clipped to zero, since $p_d \geq 0$ by physical interpretation. This is a post-processing step and does not enter the training loss.

a) *P_{50} versus L^2 point forecast: a methodological note*: The conditional median forecast ($\alpha = 0.5$) is not in general equal to the L^2 -loss-trained point forecast from notebook 05a, because bus-level p_d has a right-skewed conditional distribution (many small buses with occasional spikes, a few large buses with high baseline). On our test set, P_{50} diverges from the L^2 -trained point forecast by -11.8% at the next-month horizon (P_{50} predicts lower than mean). We surface this in Section V as a feature of the distribution, not a bug of the model.

V. RESULTS

A. Evaluation framework

All ten model variants are evaluated against the 2025 calendar year at hourly resolution for both next-day and next-month tasks. Following the discipline of Hong & Fan [6] we report MAE and Weighted MAPE (WMAPE) alongside RMSE, since the L^2 -trained models can excel at squared-error metrics while degrading on tail or peak hours. WMAPE is the assignment’s primary specified metric and is reported in all stratified results below; we additionally surface RMSE because the MinT framework is mathematically defined to minimize squared error and aggregate RMSE is the natural metric for comparing reconciliation strategies.

Bus-level and zone-level metrics are reported separately throughout. The 24 hours of 2025-12-04 are excised from all metrics (Section II-D); the resulting test window is 8,736 hours. For the MinT-augmented evaluation, the bus universe partitions into a 3,305-bus *reconciled* subset (where MinT can be applied) and a 648-bus *fallback* subset (where MinT reduces to the base model modulo non-negativity clipping); we report metrics on both partitions and clearly label which is in view.

B. Headline metrics and the architectural-disagreement finding

Table IV reports bus-level and zone-level WMAPE for each of the seven point-forecast variants (without MinT yet) on both tasks. The single most important finding in the entire report is visible in this table.

a) *The architectures genuinely disagree about who wins.*: `global_bus_lgbm_weather` dominates at bus level (7.51% WMAPE at next-day vs 25.55% for `zone_direct_lgbm_weather`). `zone_direct_lgbm_weather` dominates at zone level (4.10% WMAPE at next-day vs 4.73% for `global_bus_lgbm_weather`). This is not a draw arising from measurement noise — the bus-level gap is 18 percentage points and the zone-level gap is consistent across tasks. This is a real architectural finding: *the model an operator should use depends on what aggregation level the operator cares about.*

The mechanism is structural rather than tuning-dependent. Zone-direct optimizes the zone-aggregate objective directly and disaggregates through fixed hour-of-day shares (Section IV-B); this produces highly accurate zone forecasts but introduces averaged-shape noise at the bus level. Global-bus optimizes bus-level predictions directly with bus identity as a categorical feature, capturing bus-specific patterns at the bus level but accumulating per-bus errors when summed to zone aggregate. Each architecture is best at the aggregation level it was designed to predict.

Table V converts this finding into the operational recommendation matrix that directly answers the assignment’s question about which method works better for which task.

b) *Three baseline observations.*: (i) `hist_avg` and `prev_year` produce bit-identical predictions for both tasks, because both are task-invariant by construction (climatology and same-hour-last-year ignore forecast horizon). (ii) `prev_recent` — last week at the same hour for next-day, the trailing 63-day hour-of-day mean for next-month — is unexpectedly strong at the bus level for next-day (MAE 1.77, WMAPE 12.57%), beating both zone-direct LightGBM variants. At the bus granularity, recent autoregressive lookup captures bus-specific behavior that zone-direct’s disaggregation smears away. (iii) The structural-growth diagnostic works as designed: `prev_year`’s residuals are sharply positive in FWES and NOTH and roughly zero elsewhere, confirming the per-zone growth gradient established in Section II-C.

C. Cold-start: the most striking architectural finding

The headline numbers in Table IV are computed over the full 4,208-bus universe and so are dominated by the 4,166 non-cold-start buses. Stratifying on cold-start status reveals an architectural divide that is roughly an order of magnitude wide. Table VI reports the cold-start penalty ratio — cold-start WMAPE divided by non-cold-start WMAPE — for the seven point-forecast variants at the next-day horizon.

The global-bus family shows a penalty of 1.05–1.19 $\times$ — essentially no degradation on unseen buses. The zone-direct

family and the static sNaïve baselines show penalties of 7.6–8.8 $\times$ with cold-start WMAPE exceeding 220%. The $\sim 220\%$ figure means predicted load is roughly 2–3 $\times$ the actual load, every hour, on every cold-start bus.

The mechanism is direct. Zone-direct’s disaggregation (Section IV-B) assigns every cold-start bus the zone-mean hour-of-day share, then multiplies by the zone forecast. Cold-start buses are systematically smaller than the average bus in their zone (typically new commissioning loads or small substations), and the mean-share fallback massively overpredicts their load. The identical 220.39% figure for `hist_avg` and `prev_year` reveals that notebook 03 implemented the same zone-mean fallback for sNaïve baselines on unseen buses, producing literally bit-identical predictions across these two models on the cold-start subset.

Global-bus’s LightGBM categorical routing handles unseen `bus_unique_id` values by routing down the default direction at each split (effectively missing-value semantics), but the model still has full access to `zone_name`, calendar features, weather features, and bus-level lag features computed from the bus’s own 2025 history. The prediction reduces to “what would an average bus in this zone, at this hour, under these conditions do” — a reasonable predictor even without the specific bus’s training history.

This is the operationally most important finding in the report. A production system that ingests new buses regularly — which all real grids do, since substation commissioning is continuous — using top-down disaggregation as the bus-level forecast strategy will produce 200%+ WMAPE on every newly-commissioned bus from the moment it appears until at least one full year of training history exists.

D. Weather: a 2 $\times$ 2 decomposition by architecture and aggregation level

Adding the five-feature zonal weather set (Section III-D) improves every aggregate metric on every model on every task. The magnitude depends sharply on architecture and aggregation level. Table VII reports the relative WMAPE reduction for each of the eight combinations.

a) *Each architecture exploits weather best at its native optimization target.*: Zone-direct trains against the zone-aggregate objective, so weather features directly improve zone-level predictions (31–50% zone lift) but add only modest bus-level lift (2–7%). Global-bus trains against bus-level p_d , so weather features improve bus-level predictions directly (15% next-day) with moderate zone-level pass-through (19–25%). The biggest single result is zone-direct next-month zone-level WMAPE dropping by half (9.53% $\rightarrow$ 4.75%, lift +50.2%). Cutting WMAPE in half on a major aggregation metric is the cleanest single argument for the weather feature engineering effort.

b) *Weather is regime-dependent and FWES is the counter-example.*: Adding weather features makes FWES worse in 6 of 8 (architecture $\times$ granularity $\times$ task) cells. Table VIII shows the per-cell weather effect in FWES specifically.

TABLE IV
HEADLINE BUS-LEVEL AND ZONE-LEVEL WMAPE ON THE 2025 TEST SET. BEST IN COLUMN **BOLD**.

Model	Bus-level WMAPE		Zone-level WMAPE	
	nextday	nextmonth	nextday	nextmonth
hist_avg	29.22%	29.22%	16.14%	16.14%
prev_year	27.48%	27.48%	10.85%	10.85%
prev_recent	12.57%	25.45%	10.62%	17.11%
zone_direct_lgbm	26.20%	27.22%	6.00%	9.53%
zone_direct_lgbm_weather	25.55%	25.29%	4.10%	4.75%
global_bus_lgbm	8.84%	23.30%	6.30%	10.13%
global_bus_lgbm_weather	7.51%	22.35%	4.73%	8.20%

TABLE V
OPERATIONAL RECOMMENDATION MATRIX.

Operational requirement	Recommended model
Next-day, bus-level accuracy	global_bus_lgbm_weather
Next-day, zone-level accuracy	zone_direct_lgbm_weather, narrowly
Next-month, bus-level accuracy	global_bus_lgbm_weather
Next-month, zone-level accuracy	zone_direct_lgbm_weather, decisively all 8 zones
Grid with entity churn (cold-start)	global_bus_lgbm_weather (see Section V-C)
Zone-level + coherence required	mint_shrink (see Section V-H)

TABLE VI
COLD-START PENALTY (NEXT-DAY, BUS-LEVEL WMAPE).

Model	non-CS	CS	Penalty
global_bus_lgbm	8.84%	9.32%	1.05×
global_bus_lgbm_weather	7.51%	8.97%	1.19×
prev_recent	12.56%	25.60%	2.04×
hist_avg	29.07%	220.39%	7.58×
prev_year	27.34%	220.39%	8.06×
zone_direct_lgbm	26.05%	223.87%	8.59×
zone_direct_lgbm_weather	25.40%	223.93%	8.82×

This is the empirical confirmation of the Safdarian argument [5] in Section III-D: temperature-load correlation is near-zero in the Permian Basin industrial-dominated zone, and adding weather features there introduces noise rather than signal. The single exception (zone-direct next-month zone-level +38.4%) shows that weather signal does exist at the FWES zone aggregate over long horizons — but at the bus level or short horizons, FWES doesn’t benefit. The report’s honest framing is therefore: *weather helps where weather drives load* (metropolitan AC-driven zones), and may hurt where it doesn’t (industrial growth zones).

c) *Weather lift is much smaller for cold-start buses.*: For zone-direct, weather provides essentially zero cold-start lift (−0.95% to −0.03%). The share-fallback failure dominates

TABLE VII
WEATHER LIFT: RELATIVE WMAPE REDUCTION FROM ADDING THE FIVE-FEATURE ZONAL WEATHER SET, ALL POSITIVE.

Architecture	Task	Bus	Zone
zone-direct	next-day	+2.5%	+31.6%
zone-direct	next-month	+7.1%	+50.2%
global-bus	next-day	+15.0%	+24.9%
global-bus	next-month	+4.1%	+19.0%

TABLE VIII
WEATHER LIFT IN FWES (NEGATIVE NUMBERS INDICATE WEATHER HURTS).

Architecture	Task	Bus	Zone
zone-direct	next-day	−1.4%	−10.5%
zone-direct	next-month	+0.3%	+38.4%
global-bus	next-day	−7.7%	−7.9%
global-bus	next-month	−1.0%	−7.3%

the error so completely that weather features cannot propagate to noticeable bus-level improvement. For global-bus, weather provides 0.14–3.75% cold-start lift — modestly beneficial but well below the 4–15% non-cold-start lift. This compounds the cold-start architectural advantage from Section V-C: for production systems with entity churn, the combination of global-bus’s superior cold-start robustness *and* weather’s modest cold-start lift makes `global_bus_lgbm_weather` the clearly preferred choice for unseen buses.

E. Per-zone stratification: where each architecture wins

The headline architectural disagreement in Section V-B is directionally correct but not uniform across zones. Two zone-specific anomalies are worth surfacing:

a) *NOTH: where sNäive beats LightGBM at the bus level.*: At the bus level in NOTH, the `prev_recent` baseline beats every LightGBM variant: 12.44% WMAPE at next-day and 29.47% at next-month, versus 23.33% and higher for the LightGBM variants in the same zone. NOTH is the smallest zone with the sparsest per-bus training data; the LightGBM models, which try to learn a generalizable function

across all buses, are outperformed in NOTH by a model that simply memorizes recent observations. This is consistent with forecasting scaling-law principles — model complexity pays off where data supports it; for thin slices, simpler is better.

b) Per-zone WMAPE variation dominates per-model variation.: Metropolitan zones (COAS, NCEN, SCEN) cluster at 6–8% bus-level next-day WMAPE and 3–4% zone-level next-day WMAPE — the best in the dataset. Semi-arid or sparse zones (WEST, NOTH) reach 10–12% bus-level next-day WMAPE, nearly $2\times$ worse. Geographic and economic structure dominates forecast difficulty more than any modeling choice. The implication for production deployment: forecast quality service-level expectations should be set per-zone, not globally.

F. Per-hour stratification: flatter than expected

WMAPE is remarkably flat across hours of day for all LightGBM models, with peak-vs-off-peak ratios in the narrow 0.94–1.13 range. ERCOT bus-level load does not have a single sharp diurnal peak — the summer pattern peaks in late afternoon (HE 17–19, cooling) while the winter pattern peaks in early morning (HE 7–9, heating), and the two largely cancel at the annual aggregate. The methodological reading: our LightGBM models achieve their overall accuracy through uniform competence across hours rather than by excelling at specific hours.

The next-month task reveals a counterintuitive pattern: WMAPE is worst in early-morning hours (HE 1–7) and best in afternoon hours (HE 13–22) across all LightGBM variants. The most likely explanation is that afternoon hours are more weather-determined and the long-horizon model benefits disproportionately from the more-predictable hours; early-morning hours have higher variance-to-mean ratios driven by heterogeneous HVAC startup, industrial overnight schedules, and commute patterns.

G. Scaling-law fit and the next-month small-bus anomaly

Fitting per-bus RMSE against per-bus mean load on log-log axes yields the Sevlian–Rajagopal [10] scaling exponent β . For next-day, every model produces β in the conventional 0.44–0.63 range with $R^2 \in [0.53, 0.81]$, consistent with the $\beta \approx 0.5$ value reported across multiple US ISOs in the original paper. The cleanest fit is `global_bus_lgbm` at $\beta = 0.62$, $R^2 = 0.74$. Adding weather features flattens the slope (from 0.62 to 0.58 for global-bus, from 0.59 to 0.57 for zone-direct), evidence that weather provides relatively more benefit to larger metropolitan-AC-driven buses than to smaller ones.

For next-month, the global-bus family breaks the scaling-law pattern sharply. β collapses to 0.28 for `global_bus_lgbm_weather` with $R^2 = 0.40$, the worst-fitting model in the comparison. The scatter reveals the mechanism: the next-month global-bus model produces a near-constant absolute RMSE of 1–2 MW across small buses, regardless of their actual mean load — the predictions are regressing toward the population-mean target at long horizons.

TABLE IX
MINT AGGREGATE METRICS ON THE 3,305-BUS RECONCILED SUBSET.

Metric	Task	base	WLS	shrink
Bus RMSE (MW)	next-day	7.34	5.36	5.01
Bus RMSE (MW)	next-month	10.25	9.53	9.14
Zone RMSE (MW)	next-day	434.3	417.0	377.6
Zone RMSE (MW)	next-month	784.6	740.5	407.9

Mechanistically: with 4,208 bus categories, LightGBM’s default `max_bin=255` target-mean-encodes categories beyond bin 255 toward the global mean (~ 14 MW); for next-month, the model has fewer effective lag features to differentiate buses (the shortest admissible lag is roughly 60 days), so the categorical embedding does more of the per-bus discrimination work. Small-load buses with sparse training signal get pulled toward the global mean, inflating their predictions. For next-day, this effect is much milder because the short-horizon model has rich recent lag features (24h, 168h, trailing means) that provide strong per-bus signal independent of the categorical encoding.

The scaling-law analysis surfaces a finding not visible in aggregate RMSE metrics: *global-bus next-month performance is bimodal across bus sizes*. Large-bus accuracy is competitive; small-bus accuracy is systematically poor. An operator evaluating global-bus for next-month deployment should know this — aggregate metrics across all buses mask the small-bus failure mode.

H. Hierarchical reconciliation: the concentrated-wins finding

Applying MinT-WLS and MinT-shrink (Section IV-E) to the global-bus base forecasts produces aggregate improvements on the 3,305-bus reconciled subset. Table IX reports the headline aggregate numbers.

MinT-shrink wins the zone-level RMSE for both tasks, narrowly displacing `zone_direct_lgbm_weather` as the zone-level state-of-the-art on this evaluation. The next-month zone-level reduction is particularly substantial: 48% relative to the global-bus base (or 10.4% relative to the previously-dominant zone-direct + weather). This is the coherence-by-construction property paying off: reconciled bus-level forecasts sum exactly to optimal zone forecasts, which strictly dominates either independent zone-level forecasting or independent bus-level forecasting at the zone aggregate scale.

We now report three honest stratifications that materially qualify this aggregate result.

a) Finding 1: The MinT-shrink aggregate win is spatially concentrated, with NOTH driving most of it.: Per-zone next-day bus RMSE under MinT-shrink:

In six of eight zones, MinT-shrink produces metrics indistinguishable from or slightly worse than the base. Only NOTH and SCEN show meaningful improvement; FWES is made 23.4% worse. The headline “–32% bus-level RMSE” is therefore overwhelmingly NOTH-driven.

TABLE X
PER-ZONE BUS-LEVEL RMSE NEXT-DAY (MINT_RECONCILED SUBSET). Δ
IS MINT-SHRINK RELATIVE TO BASE.

Zone	base	shrink	Δ
COAS	3.33	3.36	+1.0%
EAST	1.49	1.62	+8.7%
FWES	3.59	4.43	+23.4% (worse)
NCEN	5.38	5.62	+4.5%
NOTH	24.60	10.86	−55.9%
SCEN	5.85	5.07	−13.3%
SOUT	2.88	2.87	−0.3%
WEST	1.09	1.09	0.0%

b) Finding 2: 35–45% of reconciled buses are hurt by MinT-shrink, not helped.: A per-bus tally on the 3,305-bus reconciled subset shows that MinT-shrink improves 2,136 buses (64.6%) but hurts 1,169 buses (35.4%) at next-day. At next-month the split is closer: 1,806 helped (54.6%), 1,499 hurt (45.4%). Mean per-bus RMSE delta at next-month is +0.157 MW (*worse* on average); the median per-bus delta is −0.058 MW (*better* at the median). MinT-shrink helps the median bus and hurts the mean bus — the classic mean-vs-median divergence indicating a heavy-tailed distribution of per-bus effects where a few large wins offset many small losses (in L^2) but not in L^1 .

c) Finding 3: A single 345 kV transmission bus drives the aggregate gain.: Per-bus inspection identifies CHLDDATA_345KV_1 (NOTH zone, 345 kV transmission bus, mean actual load 495 MW) as the largest contributor. Its per-bus RMSE drops from 345.74 MW to 151.81 MW at next-day (a −193.93 MW improvement) and from 394.75 MW to 173.25 MW at next-month (−221.50 MW). The second-most-helped bus contributes roughly 10× less in absolute MW. Estimating: NOTH has ~250 reconciled buses, so CHLDDATA’s ~194 MW per-bus improvement translates to roughly 12 MW of NOTH-aggregate RMSE reduction on its own — comparable to the entire 13.7 MW NOTH RMSE drop observed in Table X. Within NOTH at next-day, only 12 buses are helped by MinT-shrink while 163 are hurt; CHLDDATA’s 194 MW correction outweighs the 163 small regressions in aggregate RMSE.

This is not a failure of MinT — CHLDDATA is exactly the bus the theory predicts will benefit: a 345 kV transmission bus where the base global-bus model has large residual variance and the zone-level signal is more accurate than the bus-level signal, so the projection toward the zone forecast produces a large correction. But the honest framing for the report differs sharply from the aggregate headline. We adopt the latter framing:

MinT-shrink’s aggregate RMSE improvements concentrate at a small number of high-voltage transmission buses where the base global-bus model produced large errors. On the median bus, MinT-shrink produces small effects, and 35–45% of reconciled buses see RMSE regression rather than improvement. The

aggregate win at zone level is real — the coherence-by-construction guarantee mathematically improves zone aggregates — but the bus-level mechanism is concentrated rather than uniform.

d) prev_recent narrowly beats MinT-shrink at bus-level RMSE.: Once MinT is in the comparison, the bus-level RMSE ranking on the reconciled subset has *prev_recent* at rank 1 (4.96 MW next-day, 8.65 MW next-month) and *mint_shrink* at rank 2 (5.01 and 9.14). The naive last-week-lookup baseline narrowly beats every machine-learning model at this specific bus-level RMSE metric. The trade-off is at zone level, where *prev_recent*’s RMSE is 3.2× and 4.5× worse than MinT-shrink’s: per-bus week-over-week prediction errors are roughly random in direction (low bus-level RMSE) but don’t cancel when summed to zone aggregate. If last Tuesday at 3 PM had a heat wave and this Tuesday doesn’t, every bus misses in the same direction. *prev_recent* and *mint_shrink* represent the two extremes of the bus-vs-zone trade-off: one is implicitly optimized for per-bus point accuracy and wins bus-level; the other is mathematically optimized for coherent zone aggregation and wins zone-level. Neither wins both.

e) WLS vs Shrink is close at the per-bus level.: On the 3,305-bus reconciled subset, MinT-shrink wins more often than MinT-WLS (64.4% vs 35.6% at next-day, 54.3% vs 45.7% at next-month). Aggregate RMSE favors Shrink because Shrink’s wins are larger in magnitude than its losses (driven by the same CHLDDATA-class corrections), not because Shrink wins more often. Per-zone, the WLS-vs-Shrink picture is genuinely mixed at next-month: WLS wins NOTH, COAS, and FWES at per-bus count; Shrink wins NCEN and EAST. A production deployment would benefit from a per-bus or per-zone estimator selection layer rather than a single global choice between WLS and shrinkage.

f) MinT cannot help cold-start.: By construction, MinT covariance is undefined for buses without training residuals. The 42 cold-start buses are entirely in the 648-bus fallback subset; MinT predictions for these buses are bit-identical to base global-bus predictions modulo non-negativity clipping. The cold-start architectural finding from Section V-C therefore propagates through to the MinT-augmented evaluation unchanged.

I. Negative predictions and the free physical guarantee

Only the global-bus family produces negative predictions, and only at the next-day horizon. The headline rates: 0.871% negatives for *global_bus_lgbm* and 2.107% for *global_bus_lgbm_weather*. Adding weather features more than doubles the negative rate, evidence that adding strong predictive features increases the model’s willingness to extrapolate aggressively along their value range. Decomposing by MinT subset: among the 3,305 reconciled buses, the rate is 0.735%; among the 648 fallback buses, 13.197% — more than one in eight next-day predictions on the population MinT structurally cannot help.

The 167 *zero_load_2025* buses (Section IV-E, fallback breakdown) drive most of the fallback-subset negatives. These

buses have actual $p_d \approx 0$ but the model predicts ~ 0.06 MW at next-day; small noise pushes the prediction below zero frequently.

a) *MinT enforces non-negativity end-to-end.*: Notebook 07’s clipping policy was verified bit-exact in our fallback sanity check: on the 648 fallback buses, MinT’s predictions equal $\max(\text{base raw}, 0)$ for 100% of rows; the 0% negative prediction rate of both MinT variants holds across all populations. This is a small but real benefit of the MinT pipeline beyond the reconciliation itself: even where reconciliation does not apply, MinT enforces a physical constraint the base model lacks at no cost ($\max(\cdot, 0)$ is monotone).

J. Probabilistic forecasts via quantile regression

The quantile-regression LightGBM extension (notebook 08, Section IV-F) produces P_{10}, P_{50}, P_{90} prediction intervals at both horizons. Empirical observations:

- **Quantile crossing** (e.g., $\hat{P}_{10} > \hat{P}_{50}$ on some rows) before row-wise sorting: 2.01% at next-day and 0.03% at next-month. The next-day rate is non-trivial but not large enough to suggest fundamental model misspecification; sorting repairs it cleanly.
- **P_{50} vs L^2 point forecast divergence** on the test set: -11.8% at the next-month horizon (the conditional median forecast is systematically lower than the mean-trained point forecast). This is a feature of the right-skewed bus-level p_d distribution rather than a model defect: many small buses with occasional spikes produce a distribution whose median sits below its mean, and quantile regression at $\alpha = 0.5$ correctly tracks the median while the L^2 -trained model tracks the mean.
- **Calibration verification was outside scope.** We compute P_{10} and P_{90} predictions and apply zero-clipping to the lower tail, but we do not formally verify empirical coverage (PICP) or sharpness (PINAW) against the 2025 ground truth. We document this as a methodological honesty disclosure in Section VI rather than reporting unverified calibration metrics.

The quantile extension is therefore a partial deliverable. The infrastructure (six trained quantile models, six deliverable forecast parquets) is complete and committed to the repository; the empirical verification of the calibration is deferred.

VI. DISCUSSION AND LIMITATIONS

This section airs four methodological honesty disclosures the careful reader should be aware of when interpreting the results, then identifies six concrete future-work directions each anchored in a specific literature precedent and a specific result from Section V.

A. Honesty disclosure 1: strict warm-start trades hyperparameter optimality for clean attribution

The weather-augmented variants (notebooks 04b and 05b) reuse the hyperparameters discovered by Optuna on the base feature set (Section IV-D), with no re-tuning. This is the GEFCom 2012 controlled-feature-ablation discipline [20]: any

RMSE delta is attributable to the weather features alone, not confounded with hyperparameter shifts. The empirical cost is visible in the next-day `global_bus_lgbm_weather` negative-prediction rate of 2.107% (Section V-I), more than $2\times$ the rate of the no-weather base variant (0.871%). The added weather features expand the directions in which LightGBM trees can extrapolate, but the hyperparameters — particularly `min_data_in_leaf` and `cat_smooth` — were not re-tuned to regulate the new feature space. A re-tuned variant would likely produce fewer negatives, possibly at the cost of slightly weaker non-negative predictions.

We accept this cost because the alternative confounds two effects (the contribution of weather features vs. the contribution of hyperparameter retuning) and would not let us attribute the $+15\%$ next-day bus-level weather lift cleanly to weather features themselves. The cleanest follow-up experiment is a second-round Optuna search on the weather-augmented feature set, which would either confirm or reverse the strict-warm-start finding. We did not run this experiment within scope.

B. Honesty disclosure 2: MinT covariance is estimated on the test set (Option Y)

The MinT framework requires an estimate of the base-forecast error covariance matrix W . We estimate W on the residuals of `global_bus_lgbm_weather` and `zone_direct_lgbm_weather` against the 2025 test ground truth, with shrinkage intensity λ chosen by the Schäfer-Strimmer optimization criterion [11] on the same residuals (Option Y). This is methodologically the weakest practical choice: a production system would estimate W on a held-out subset of 2024 validation residuals (Option Z) to eliminate in-sample tuning of the reconciliation step.

We adopt Option Y because partitioning 2024 to create a covariance estimation subset would have required either (a) reducing the training-window size for the base forecasts (potentially confounding the MinT comparison with weaker base models) or (b) running a separate covariance-estimation pass against a 2024 sub-window with its own forecast set — substantial engineering with limited scope return. The empirical shrinkage intensity ($\lambda \approx 0.002\text{--}0.005$) suggests we are in the favorable $N \ll T$ regime where the full sample covariance is well-conditioned and shrinkage provides only marginal benefit beyond the diagonal; the methodological risk of Option Y is correspondingly small but non-zero. A production deployment would re-estimate W properly.

C. Honesty disclosure 3: the MinT aggregate wins are concentrated, not uniform

We restate the Section V-H headline for emphasis. MinT-shrink’s aggregate bus-level RMSE improvement of -32% at next-day on the 3,305-bus reconciled subset is overwhelmingly driven by a single 345 kV transmission bus (`CHLDDATA_345KV_1`) whose individual RMSE drops by ~ 194 MW. NOTH as a whole shows only 12 buses helped versus 163 hurt by MinT-shrink at next-day; the zone-level aggregate improvement exists only because CHLDDATA’s

single correction outweighs the 163 small regressions. Across the full reconciled subset, 35.4% of buses are hurt at next-day and 45.4% at next-month. The mean per-bus RMSE delta at next-month is *positive* (i.e., worse on average per bus) even though the zone-aggregate RMSE improves substantially.

This is not a defect of MinT — it is the predicted behavior of an L^2 -minimizing reconciliation against base forecasts with a heavy-tailed residual distribution. Large per-bus residuals at high-voltage transmission buses dominate the aggregate L^2 objective and are exactly the residuals MinT can most usefully correct. But the honest framing matters: a grid operator should not interpret “MinT reduces zone-level RMSE by 10% on average” as “MinT improves each individual bus’s forecast.” The improvement is concentrated at transmission-tier buses, while most buses see either neutral or modestly negative effects. The deployment implication is that an operator running bus-level distribution analytics on typical-sized buses might rationally prefer the unreconciled global-bus base; the operator running ISO-level dispatch on zone aggregates should prefer MinT-shrink.

D. Honesty disclosure 4: probabilistic forecasts produced without empirical calibration verification

Notebook 08 trains and persists quantile-regression LightGBM models at $\alpha \in \{0.1, 0.5, 0.9\}$ for both tasks, six deliverable quantile-forecast parquets in total. We document the quantile-crossing rates (2.01% next-day, 0.03% next-month before row-wise sorting) and the P_{50} vs L^2 point-forecast divergence (−11.8% at next-month). What we do *not* report is formal calibration metrics — Prediction Interval Coverage Probability (PICP), Prediction Interval Normalized Average Width (PINAW), or pinball-loss decomposition against the 2025 ground truth. The infrastructure to compute these exists in the repository; the empirical verification step was outside scope.

The honest framing is that notebook 08 is a partial deliverable: the model infrastructure and forecast files are complete and committed; the calibration verification is the missing piece. A reader who needs calibrated prediction intervals should treat our P_{10} and P_{90} outputs as provisional pending empirical coverage verification.

E. Future work, anchored to concrete results

We identify six directions each motivated by a specific result in Section V and supported by a specific literature precedent.

a) 1. Grouped-Global-Bus clustering for next-month small-bus bias: The scaling-law finding (Section V-G) revealed that `global_bus_lgbm_weather` produces a near-constant $\sim 1\text{--}2$ MW RMSE on small next-month buses regardless of their actual mean load, driven by the categorical encoding of `bus_unique_id` pulling rare bus identities toward the ~ 14 MW global mean. The Grouped-Global-Bus variant of Triebe et al. [3] addresses this by training a separate global model per cluster of similar buses. Clustering by `tsfeatures` signatures (autocorrelation profile, seasonality strength, trend strength) on the 2022–2024 training window would produce $k \in \{3, 5, 8\}$ size-and-shape-homogeneous bus

clusters, each trained as its own global model. Triebe et al. report this yields a 5% RMSE reduction at the utility aggregate over the pure Global-Bus variant; the volatility-heterogeneity tradeoff at our 4,208-bus scale would need to be measured empirically.

b) 2. Bus-specific weather joins via Voronoi catchment: The substation-level weather-coupling heterogeneity argument (Section III-D, Argument 1) and the FWES weather-resistance finding (Section V-D) both point to the same limitation: a single airport station per zone averages over buses whose true weather-coupling spans nearly two orders of magnitude. Schröter et al. [4] report substation-level temperature–load correlations spanning $r = 0.016$ to $r = 0.498$ within the same 50Hertz TSO. The principled fix is a bus-specific weather join: assign each bus to its nearest weather station via Voronoi tessellation on geographic coordinates, then engineer the same five features per bus rather than per zone. This would materialize $\sim 50\text{--}100$ weather feature columns per bus rather than collapsing them all to a single zone average, and would directly test whether FWES’s weather-resistance is a zone-level artifact or a genuine industrial-load property of its constituent buses.

c) 3. Peak-weighted loss for rare-event hours: Winter Storm Elliott (Section II-D) is the only extreme cold event in our 2022–2024 training window and one of two single-event peak regimes in our test set. An event-flag feature lets LightGBM learn an offset for the storm hours, but with $n = 1$ training exemplar the learned offset is necessarily high-variance. Mathew et al. [16] report the PWP (Peak-Weighted Pinball) design pattern: a quantile-pinball loss with sample weights inversely proportional to the empirical quantile rank of each hour, so peak hours contribute disproportionately to the gradient. PWP generalizes across *all* peak hours rather than relying on a single rare-event flag, and converges to standard quantile regression as the weight ratio $\rightarrow 1$. Implementing PWP as a LightGBM `objective='custom'` would be roughly a half-day of engineering and would directly target the bus-vs-zone aggregation tradeoff at high-stake hours.

d) 4. Proper MinT covariance on held-out 2024 (Option Z): Honesty disclosure 2 (above) identifies the cleanest improvement available to the MinT pipeline. Reserving the last six weeks of 2024 as a held-out covariance estimation window, refitting the base forecasts on $\{2022, 2023, 2024 \text{ minus held-out}\}$, and re-running the MinT projection with the held-out residuals as W ’s training data would produce an Option-Z-compliant MinT result. The expected effect on aggregate metrics is small (because the empirical λ is small), but the methodological claim would be substantially cleaner. This is $\sim 2\text{--}3$ days of engineering.

e) 5. Empirical calibration verification for quantile forecasts (PICP/PINAW + reliability diagrams): Honesty disclosure 4 (above) identifies the gap. The verification is arithmetic — compute the fraction of 2025 actuals falling within $[\hat{P}_{10}, \hat{P}_{90}]$ per (model, task, zone), compare to the nominal 80% coverage, plot a reliability diagram across quantile deciles. This is ~ 1 day of engineering and would convert notebook 08 from a partial

deliverable into a complete one. The likely finding given the right-skewed p_d distribution is that the P_{90} upper bound is well-calibrated (large buses with summer peaks live near the conditional-90th percentile) while the P_{10} lower bound over-covers (the lower tail clips at zero by physics, compressing the calibrated lower quantile).

f) *6. Cold-start meta-learner to replace the share fallback:* The cold-start finding (Section V-C) revealed that zone-direct’s 200%+ WMAPE on unseen buses is a structural disaggregation issue that weather features cannot fix. The principled replacement: train a meta-learner predicting initial bus share from (*zone, hour, base_kv, calendar features*) on the subset of *historical* cold-start buses — buses that appeared in 2023 or 2024 for the first time. This converts cold-start from a fallback problem to a forecasting problem in its own right, learnable from data. The expected effect is to close most of the 200%+ WMAPE gap and bring zone-direct’s cold-start handling closer to global-bus’s $\sim 10\%$ WMAPE on the same population. This is $\sim 3\text{--}5$ days of engineering and would be the most operationally valuable single addition to the pipeline.

VII. REPRODUCIBILITY

The complete pipeline — twelve Jupyter notebooks, twenty forecast files (canonical seven-column schema), all Optuna hyperparameter artifacts, MinT diagnostic outputs, and evaluation metric tables — is committed to a public Git repository:

<https://github.com/GavinGongTech/ECESIS-2026-Summer-Power-Systems-Modeling-Assignment>

a) *Quick-start for users:* The repository uses Git LFS for the forecast deliverables (~ 2.3 GB in 20 parquet files) and small artifacts (~ 50 MB across model parameter JSONs, evaluation metric parquets, and MinT diagnostic outputs). You can clone the repo using standard practice.

The forecast deliverables are then available under `assignment2/data/processed/forecasts/`. A Python environment specification (`environment.yml`) is provided; the only non-standard dependencies are `lightgbm`, `optuna`, `meteostat`, and `pyarrow`.

b) *Repository structure follows a four-tier git policy:*

- **Raw inputs** (gitignored, source: Google Drive distribution from the assignment). Bus-level and zone-level parquets, 2022–2025. Documented in `README.md` with the original distribution link.
- **Intermediate features** (gitignored, regenerable via notebook 02 and 02w). The eight feature-matrix parquets (~ 2 GB) and the weather feature parquet (~ 2 MB). Skipping these keeps the repository small; running notebooks 02 and 02w regenerates them deterministically from the raw inputs.
- **Forecast deliverables** (LFS-committed). The twenty seven-column forecast parquets specified by the assignment.
- **Final artifacts** (committed plain, not via LFS). Optuna best-hyperparameter JSONs, evaluation metric tables, MinT diagnostic outputs, AI usage logs, this report’s source. Small enough to commit directly.

c) *Notebook inventory:* The twelve notebooks execute in dependency order:

- 01_eda_and_data_audit: the 18,643 $\rightarrow$ 4,208 funnel and all data-quality artifacts.
- 02_feature_engineering, 02w_weather_features: the eight feature-matrix parquets plus the zonal weather features parquet.
- 03_snaive_baselines: the three sNaive baselines across both tasks.
- 04_zone_direct_lightGBM, 04b_zone_lightGBM_weather: zone-direct LightGBM with and without weather features.
- 05a_global_bus_lightGBM, 05b_global_bus_lightGBM_weather: global-bus direct LightGBM with and without weather features.
- 06_evaluation, 06b_evaluation_with_mint: the metric tables and stratifications in Section V.
- 07_mint_reconciliation: the MinT-WLS and MinT-shrink projections producing four deliverable forecast parquets.
- 08_quantile_regression: the quantile-regression LightGBM extension producing six deliverable forecast parquets.

End-to-end runtime on a single 2024-vintage laptop CPU (no GPU) is approximately 6 hours; notebook 05a (global-bus Optuna search) dominates the wall-clock at ~ 2.4 hours total across the two tasks.

d) *AI usage log:* This work used Claude (Anthropic) for literature review, methodological sounding, and drafting assistance. The `assignment2/ai_usage/` directory contains a structured log identifying the sessions, the role of AI in each, and a 4-page synthesis document explaining where AI input shaped methodological choices (notably: the seven-piece literature evidence cluster for the regime-dependent weather contribution in Section III-D and the concentrated-wins honest framing revision of the MinT results in Section V-H). All code, evaluation, and final methodological decisions were authored and verified by the human author; AI was used as a literature-search and drafting tool, not as a delegated decision-maker.

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